2021

Visualization and Statistical Analysis of CAR Data

AN ANALYSIS OF THE CAR MARKET OF BMW SERIES 1 IN, UK

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EXECUTIVE SUMMARY

This is a business study prepared for a market research organisation Car4All to investigate the used BMW 1 series class automotive market in the west midlands area B235YW within the region of 30 miles. A dataset covering numerous automotive statistics such as price, age, mileage, engine power, engine size, transmission and fuel type was contributed by the used vehicle website Autotrader.com. Following the pre-processing of the data, a suitable sampling strategy is used to obtain a sample from the population.

All of the visuals are constructed using theoretical ideas learned in class, and conclusions are drawn from each graph. Increase in the age of cars results in the decrease in its price , high age value cars have less mileage. Many statistical variables are calculated in order to interpret the sample. We utilise a confidence interval to estimate the mean of our population, and we do various hypothesis tests to examine if the different automotive qualities are dependent on each other, as well as to compare the average price of our sample to the average price of 1 Series cars in the UK.

The correlation matrix is displayed to discover which elements have a high impact on the pricing and which have a low impact. Finally, a car price calculator is developed from an appropriate parsimonious model utilising linear regression analysis.

DATA COLLECTION

To estimate the pricing market for a specific car(BMW Series1) in the second hand car industry looking at various aspects and properties related to the make and usage of cars like engine power, gearbox, registration, colour, fuel type etc. This is foe intended usage for predicting car prices for a company called Car4All. We used data from the website https://www.autotrader.co.uk/ for determining the important factor that contribute to the price of the second hand BMW series1 in pin code B23 5YW within the 30 miles radius from the year 2016-2020.

The data is pulled within the limit of 30miles radius, of pin code B23 5YW analysing one make of car model which is BMW series1. This limits the results of the analysis in the following ways:

a- the results may not be applicable to a different pin code
b- results are for one specific car make and not be applicable for different models
c- results are for old cars not exceeding 5 years and may not be applicable for any older cars
d- Outliers in the population are searched for because they vary from our statistical analysis
e- Few automobile attributes aren't included since data isn't accessible for all of them.

Tools used for statistical analysis

- Excel is used for visualization
- Python (descriptive statistics, confidence interval hypothesis, correlation, regression)

SAMPLING METHOD AND REASON FOR 5 YEAR RANGE

To make the sample representative of the population we wanted to include all types and makes of the single model analysed. Hence we have utilized Simple Random Sampling as the data for pricing had a lot of differentiating categorical variables which needed to be represented properly. In simple random sample each data have an equal and fair chance of being selected, which will resulted into unbiased and unaffected selection of the data.
The automobiles in the dataset is for the purpose of this research varied from 2016 to 2020, in order to cover a relatively current pricing range and to identify cars that would largely be compatible with clean air zones in the West Midlands area.

SAMPLE DATASET VIEW

	price	age	mileage	engine_size	engine_power	transmission	fuel
0	14300	2	15745	1.0	125	Manual	Petrol
1	17280	2	14220	1.0	140	Manual	Petrol
2	16300	2	10154	1.0	125	Manual	Petrol
3	17000	2	13393	1.0	125	Manual	Petrol
4	14999	3	27750	1.0	100	Manual	Petrol

DESCRIPTIVE STATISTICS OF SAMPLE

	price	age	mileage	engine_size	engine_power
count	97.000000	97.000000	97.000000	97.000000	97.000000
mean	15257.886598	3.195876	30221.494845	1.258763	127.391753
std	3490.276472	1.221672	20593.909154	0.347836	21.674946
min	9250.000000	1.000000	6028.000000	1.000000	94.000000
25%	12925.000000	2.000000	15189.000000	1.000000	123.000000
50%	15195.000000	3.000000	29000.000000	1.000000	125.000000
75%	17200.000000	4.000000	41333.000000	1.500000	136.000000
max	27800.000000	5.000000	120480.000000	2.000000	190.000000

Table1 : Descriptive Statistic Of Sample

The table shown above presents descriptive statistics of the sample dataset and the statistic shown in the table are explained below:

- **Count** – Number of the dataset. There are 97 cars in the dataset.
- **Mean -** The average value of a measure which other values are spread out. The table shows the average price of the BMW series1in the dataset to be £15257.88 with an average of 30221.49 miles.
- **Median** - The central value of the dataset.
- **Standard Deviation** -The measure of the dispersion of the values in the dataset relative to the average value. A high standard deviation indicates a wide spread in the values of the dataset and vice versa.
- **Minimum -**The lowest value of a particular measure in the dataset. In the dataset above the minimum price of the car BMW series 1 is £9,250
- **Maximum -** The highest value of a particular measure in the dataset. In the dataset above the maximum price of the car BMW series 1 is £27,800

PIVOT TABLE

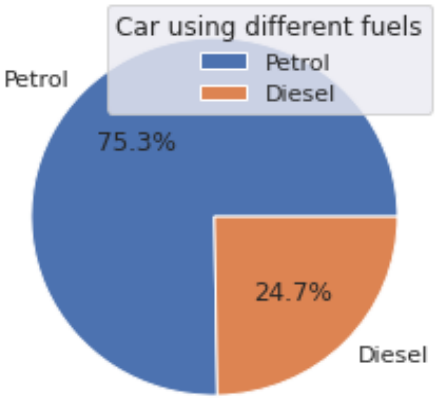
The pivot table displayed below shows the cars in the dataset grouped by fuel type (petrol and diesel) and transmission. The engine size of the cars in the dataset were distinctly 1.0L (small) and 1.5L (medium) and 2L is (large). The mileage, average price, engine size and engine power of the cars according to the specified grouping is shown

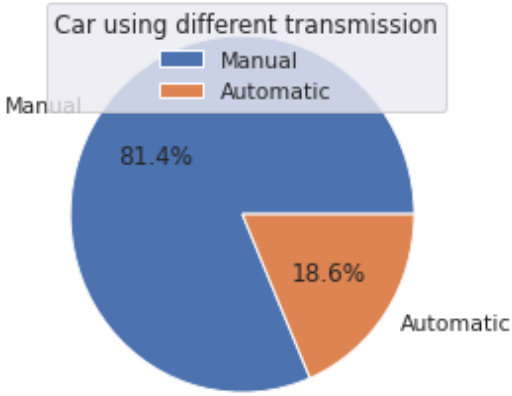
fuel	transmission	mileage	price	engine size	engine power
diesel	Automatic	42567	17467.7	1.92	168
	Manual	22833.23	13193.4	1.5	115.05
petrol	Automatic	30609.09	17584.1	1.4	138.18
	Manual	25300.82	15161.5	1	124.27

From the table above it is specified that diesel cars with automatic transmission with the large engine size of (1.92L) had the highest average price of £17,467 while the manual cars manual cars with the low engine size (1.5L) had the lowest average price of £13,193. Overall automatic cars on petrol had the highest overall average value of £17584.

VISUALIZATION OF GRAPHS

Graphical representation of the car percentages on fuels and transmission

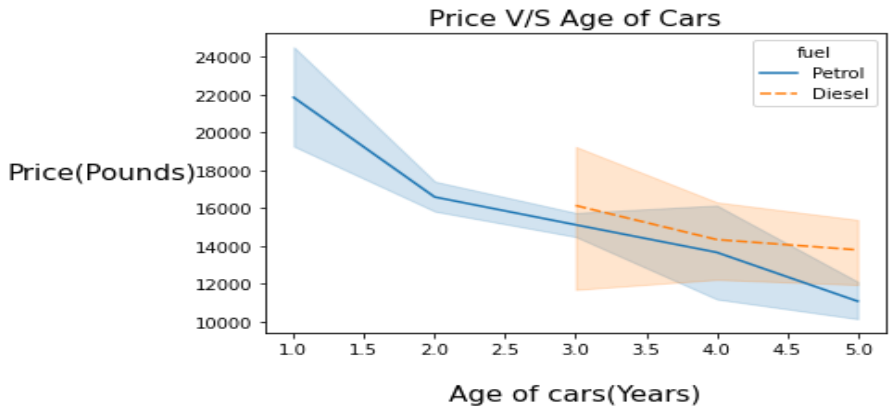




- Pie charts are helpful for showing the relationship of the parts
- The pie chart gives the numerical proportion of cars on the basis of different data fuels and transmission mode.
- The percentage of petrol cars is greater than the percentage of diesel cars for sale.
- Also, in the following chart, the percentage of manual cars is greater than the percentage of automatic cars.
- The graph follows the Gestalt principles of:
- Proximity: Data labels are close to the bars they represent
- Similarity: Number of bedrooms are shown by similar colours in the chart
- Proportionality: The graph is proportional as y-axis includes zero and scale is linear
- Clarity: Graph is well labelled
- The number of dimensions is appropriate, and the graph shows data variation

LINE GRAPH

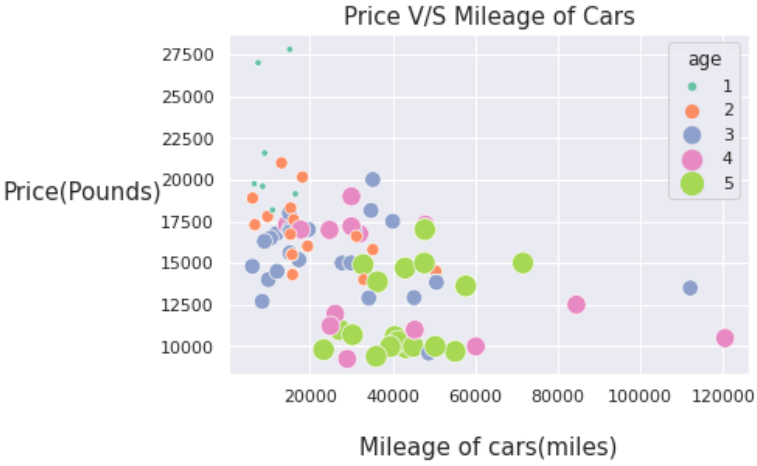
Trend between Fuel and Age of car with respect to its price



- The line is used to track changes for the period of time
- The above line graph trend of price with respect to age of car.
- With the increase in the age of cars results into the price of cars decreases.
- The price of Petrol cars is higher than Diesel cars..
- Highest price of a petrol car is £22000 whereas the highest price of a Diesel car is around £16000. Major differences can be observed in the price of cars using different fuels. But both car types follow the decreasing trend in price with respect to age.
- The graph follows the Gestalt principles of proximity, similarity, clarity, contextuality and continuity.
- Proportionality : The graph is not proportional as y- axis not include zero but the scale is linear
- The use of white background and distinct colours for age of car follows Gestalt's principle of figure/ground
- This is a multivariate plot consisting of price, age and fuel. So, this follows principles of Tufte's Graphical excellence
- As the graph represents the trend between Fuel and Age of car with respect to its price with no overlapping, it follows the MECE rule (Mutually Exclusive and Collectively Exhaustive)

SCATTER PLOT

comparison of price and mileage of cars with respect to their age



- The scatter plot above illustrate the comparison of price and mileage with respect to their age.
- The scatter plot was used here because it is very effective in illustrating comparisons across big data series, and the trend indicates that cars having less mileage are likely to have more price in the market. With high age value, cars have less mileage.
- The use of white background and distinct colours for age of car follows Gestalt's principle of figure/ground and that of focal point also, it satisfy the proximity, similarity, clarity, contextuality and continuity principles of Gestalt
- As the graph represents the comparison of mileage and price with respect to their age with no overlapping, it follows the MECE rule (Mutually Exclusive and Collectively Exhaustive)

CONFIDENCE INTERVAL LEVEL FOR AVERAGE CAR PRICE

Confidence interval: Indicates the certainty that the population mean will fall between a given range of values. Expressed in terms of 95% or 99% confidence usually.

age	mean	count	std	ci95_hi	ci95_lo
1	21863.857143	7	3922.447766	24769.647112	18958.067173
2	16596.304348	23	1934.296805	17386.828727	15805.779969
3	15257.870968	31	2213.151806	16036.958813	14478.783123
4	14134.125000	16	3418.212727	15809.049236	12459.200764
5	12305.650000	20	2630.304951	13458.431972	11152.868028

The result above shows the 95% confidence interval of second -hand car prices.
The interval comes out as

- Car which is 1 year old will have an average selling price between (18958.067173, 24769.647112)

- Car which is 2 year old will have an average selling price between (15805.779969, 17386.828727)
- Car which is 3 year old will have an average selling price between (14478.783123, 16036.958813)
- Car which is 4 year old will have an average selling price between (12459.200764,15809.049236)
- Car which is 5 year old will have an average selling price between (11152.868028,13458.431972)

So, there are 95% chances to find an average second hand car price in this rang

AVERAGE PRICE COMPARISON OF SAMPLE CAR WITH UK AVERAGE CARS

The average price of the BMW Series1 by year model is displayed in the table below, based on data acquired from Desperate Seller, another UK-based vehicle selling website:

YEAR	2016	2017	2018	2019	2020	TOTAL AVERAGE PRICE
AVERAGE PRICE (£)	15,002	17583	19,425	22,804	28,709	**20,000**

Hypothesis Testing : It is a technique used to test whether the conclusions derived from our sample data are in accordance with assumptions or pre-existing statements.

Confidence interval for mean prices:
The true mean value will fall between 21491.1966 and 22263.4433
Our null hypothesis is that the difference between the mean of the population and mean of our sample equals zero ,i.e., There is no difference between them.
The alternative hypothesis is that there is a difference, either high or low, between them.
In other words,

H0 : $\boldsymbol{\mu} = \boldsymbol{\mu}0$
H1 : $\boldsymbol{\mu} \neq \boldsymbol{\mu}0$

$\boldsymbol{\alpha} = 0.05$

T-Test

[DataSet3]

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
price	97	15257.89	3490.276	354.384

One-Sample Test

Test Value = 20000

	t	df	Significance One-Sided p	Significance Two-Sided p	Mean Difference	95% Confidence Interval of the Difference Lower	95% Confidence Interval of the Difference Upper
price	-13.381	96	<.001	<.001	-4742.113	-5445.56	-4038.67

One-Sample Effect Sizes

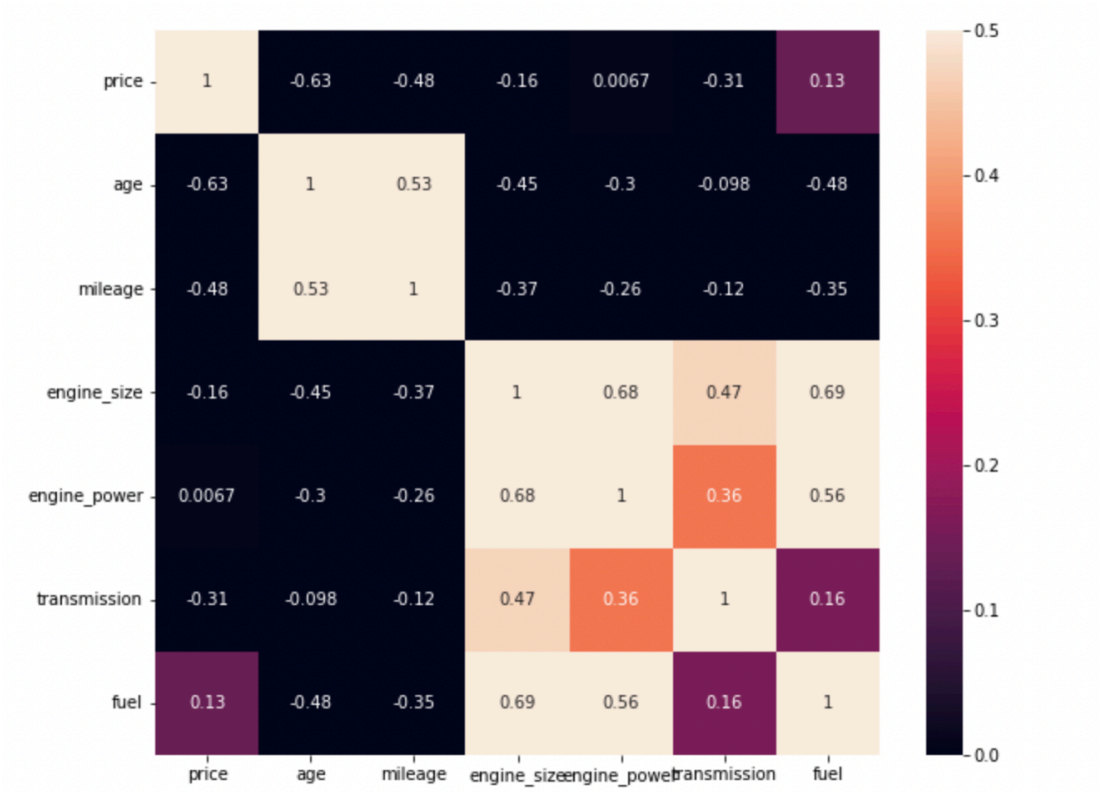
		Standardizer[a]	Point Estimate	95% Confidence Interval Lower	95% Confidence Interval Upper
price	Cohen's d	3490.276	-1.359	-1.633	-1.080
	Hedges' correction	3517.843	-1.348	-1.620	-1.072

a. The denominator used in estimating the effect sizes.
Cohen's d uses the sample standard deviation.
Hedges' correction uses the sample standard deviation, plus a correction factor.

As a result, we conducted the statistical t test and the results of the test shows that our sample mean will fall into the acceptance area. We will accept the null hypothesis that there is no difference between the mean of the population and the sample of the population

CORRELATION ANALYSIS

A correlation analysis is performed to examine how closely the variables in the dataset were connected to one another. The coefficient of correlation is the number that determines the degree of dependency between variables; coefficient values near to +1 indicate a strong positive correlation, while those close to -1 suggest a strong negative correlation. The heatmap shows the detailed results of the analysis:



Fuel is positively correlated with price with correlation coefficient more than 0.13 i.e., increase in one of these variables will increase the price.
Age, mileage, engine size, transmission is negatively correlated with price as they are inversely proportional. Engine power are not so much correlated with price as they don't affect car price significantly. So highly correlated variables are the ones that need to be considered when predicting car price
.

REGRESSION ANALYSIS

A regression analysis was performed on the dataset to determine categorically which variables were significant to the price, allowing us to answer which factors actually matter the most when determining the price of a car and which can be ignored. For the purpose of this statistical report, the dependent variable is the price of the car and the independent variables are year, mileage, engine size, engine power, transmission, and fuel are the factors which we have collected for the analysis.

PARSIMONIOUS MODEL

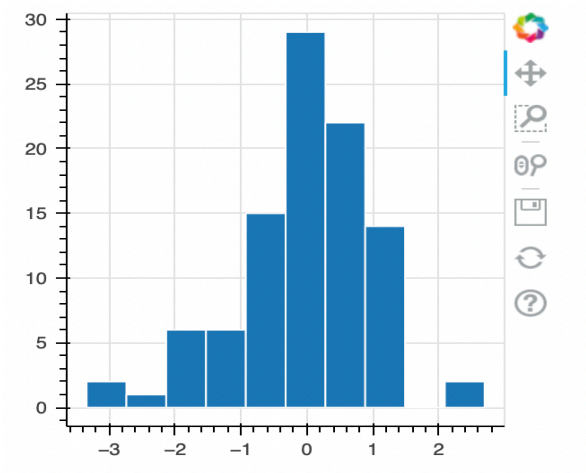
For the most parsimonious model, the independent variables were engine power of the car and fuel
A summary of the model is shown below:

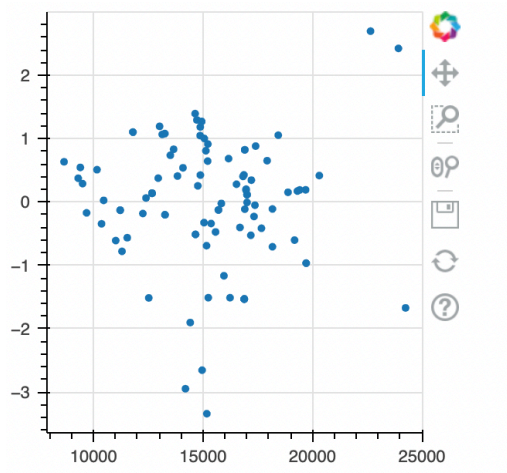
OLS Regression Results

Dep. Variable:	price	**R-squared:**	0.773
Model:	OLS	**Adj. R-squared:**	0.761
Method:	Least Squares	**F-statistic:**	62.07
Date:	Wed, 15 Dec 2021	**Prob (F-statistic):**	7.83e-28
Time:	21:06:47	**Log-Likelihood:**	-856.46
No. Observations:	97	**AIC:**	1725.
Df Residuals:	91	**BIC:**	1740.
Df Model:	5		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
Intercept	2.769e+04	846.542	32.715	0.000	2.6e+04	2.94e+04
age	-2037.8196	178.084	-11.443	0.000	-2391.562	-1684.077
mileage	-0.0538	0.010	-5.310	0.000	-0.074	-0.034
transmission	-1684.3456	511.635	-3.292	0.001	-2700.645	-668.046
engine_size	-3284.9663	386.035	-8.509	0.000	-4051.778	-2518.155
engine_power	1183.2943	347.349	3.407	0.001	493.329	1873.260

Omnibus:	12.747	**Durbin-Watson:**	1.839
Prob(Omnibus):	0.002	**Jarque-Bera (JB):**	15.549
Skew:	-0.688	**Prob(JB):**	0.000420
Kurtosis:	4.397	**Cond. No.**	1.84e+05





The assumptions of linear regression are:

1. **Linearity:** We see that the residuals are randomly scattered around zero, and there is no apparent pattern to the residuals, which suggests that a linear model is appropriate for this relationship
2. **Homoscedasticity:** The variance of residual is the same for any value of X. This isn't a time series data, so this is not violated.
3. **Independence of errors:** Observations are independent of each other. This isn't a time series data, so this is not violated.
4. **Normality:** For any fixed value of X, Y is normally distributed. This can be confirmed with the histogram above. It assumes the bell-shaped curve. A goodness-of-fit test can also be measured if sample data has skewness and kurtosis that are like a normal distribution.
5. **Multicollinearity:** From the correlation matrix between the independent variables. No two variables of my most parsimonious model is highly correlated. So, no multicollinearity issues found

SUMMARY OF THE TABLE

From the regression analysis, summary model has adjusted square value of 0.76, which the model explains efficiently by 76%. From the other independent variables, we can derive the formula for price
Y(PRICE) = 2.769e+04 -2037.8196(age) - 0.0538 - 1684.3456(transmission) - 3284.9663(engine size) + 1183.2943(engine power)
All independent variables have p value less than 0.05 it makes model good parsimonious model. We can either compare t-value or significance value (p-value) for checking good model.

CONCLUSION

The chances of prediction coming true is more than 75% so it is highly recommended to the client. Also, the national average or the sample average price of the car are very close to each other which further strengthen the model. The market trends are being followed by the model. Hence, the model can be used by the buyers.